This assignment will not be graded and is only for practice.

Level 1:

1) Match the matrices in Column A with their inverses in Column B. ***1 point***

	Column A		Column B
a)	$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 3 & 3 \end{bmatrix}$	i)	Inverse does not exist
b)	$A_2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$	ii)	$B_2 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{3} \end{bmatrix}$
c)	$A_3 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$	iii)	$B_3 = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{1}{3} \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$

Table: M2W2AQ1

- ☐ a → iii
- ☐ a → ii
- ☐ b → iii
- ☐ b → i
- ☐ c → i
- ☐ c → ii

No, the answer is incorrect.
Score: 0

Accepted Answers:

a → ii
b → iii
c → i

(Use the below information to answer questions 2 and 3).
Consider a system of linear equations

$$\begin{aligned} 2x_1 - x_2 &= 3 \\ x_1 - x_3 &= 3 \\ x_2 - x_3 &= 2 \end{aligned}$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$.

2) Choose the set of correct options. ***1 point***

- [ ] $A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$
- [ ] $A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$
- [ ] Adjoint of the matrix A is $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$
- [ ] $det(A) = 1.$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$

Adjoint of the matrix A is $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$

$det(A) = 1.$

3) Choose the set of correct options. ***1 point***

- [ ] $x_1 = -2.$
- [ ] $x_2 = 1.$
- [ ] $x_3 = -1.$
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_2 = 1.$
$x_3 = -1.$

4) Choose the set of correct options. ***1 point***

- [ ] A system of linear equations $Ax = b$ is called a homogeneous system of linear equations if $b = 0$.
- [ ] A system of linear equations $Ax = b$ is called a non-homogeneous system of linear equations if $b = 0$
- [ ] If v is a solution of the system of linear equations $Ax = b$, then $\frac{1}{c}v$ is a solution of system of linear equations $cAx = b$, where $c = 0$.
- [ ] Let $Ax = b$ be a system of linear equations. If A is invertible, then $adj(A)x = b$ also has a solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A system of linear equations $Ax = b$ is called a non-homogeneous system of linear equations if $b = 0$
If v is a solution of the system of linear equations $Ax = b$, then $\frac{1}{c}v$ is a solution of system of linear equations $cAx = b$, where $c = 0$.
Let $Ax = b$ be a system of linear equations. If A is invertible, then $adj(A)x = b$ also has a solution.

Suppose C is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Let C_{ij} be the ij-th element of the cofactor matrix C.

Answer the questions 5,6 and 7.

5) Find the value of C_{12}.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

6) Find the value of C_{22}.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) -3

1 point

7) Find the sum of the elements in the third row of C.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2:

8) Which of the following square matrices of order 3 are the same as their adjoint matrices? ***1 point***

- [ ] Identity matrix.
- [ ] Zero matrix.
- [ ] Any scalar matrix.
- [ ] Any diagonal matrix.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Identity matrix.
Zero matrix.

9) Choose the set of correct options. ***1 point***

[Hint: For the first option, find out the adjoint matrix of an arbitrary square upper triangular matrix of order 2, i.e., take the matrix $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$ and find out its adjoint matrix.]

- [ ] The adjoint of a 2×2 real upper triangular matrix is an upper triangular matrix.
- [ ] There exists a square matrix of order 3, such that $A = A^{-1} = adj(A)$.
- [ ] If $A = A^{-1}$, then $det(A)$ must be 1.
- [ ] If $A = A^{-1}$, then A must be an identity matrix.

- [ ] If $A = A^{-1}$, then $A^2 = I$.
- [ ] If $A^{-1} = adj(A)$, then $det(A)$ must be 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The adjoint of a 2×2 real upper triangular matrix is an upper triangular matrix.
There exists a square matrix of order 3, such that $A = A^{-1} = adj(A)$.
If $A = A^{-1}$, then $A^2 = I$.
If $A^{-1} = adj(A)$, then $det(A)$ must be 1.

10) If A and B are squares matrices of order 2, then choose the set of correct options. ***1 point***

[Hint: Find out the adjoint matrix of an arbitrary square matrix of order 2, i.e., take the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and find out its adjoint matrix.]

- [ ] $adj(AB) = adj(A)adj(B)$.
- [ ] $adj(AB) = adj(B)adj(A)$.
- [ ] $adj(A + B) = adj(A) + adj(B)$.
- [ ] $adj(A^T) = adj(A)^T$.
- [ ] $adj(A^{-1}) = adj(A)^{-1}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$adj(AB) = adj(B)adj(A)$.
$adj(A + B) = adj(A) + adj(B)$.
$adj(A^T) = adj(A)^T$.
$adj(A^{-1}) = adj(A)^{-1}$.

Your score is: 0/10